

The value of livestock abortion surveillance in Tanzania: identifying disease priorities and informing interventions

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Abstract

Background

Lack of reliable data on the aetiology of livestock diseases, especially in Africa, is a major factor constraining the design of effective livestock health interventions to improve livelihoods, food security and public health.

Rationale

Livestock abortion is an important disease syndrome that affects productivity and livestock economies, and poses risks to public health. Worldwide, several pathogens are associated with livestock abortions but across Africa livestock disease surveillance data rarely include information from abortion events, little is known about the aetiology and impacts of livestock abortions, and data are not available to inform prioritisation of disease interventions.

Methods

This paper describes outcomes from a surveillance platform that was set up in 15 wards in northern Tanzania spanning pastoral, agropastoral and smallholder agro-ecological systems to investigate the causes and impacts of livestock abortion. Abortion cases were reported by farmers to Ministry of Livestock and Fisheries livestock field officers and on to investigation teams. The abortion was eligible for inclusion if the research field team or livestock field

officer could attend within 72 hours of the event occurring. If so, a field investigation was carried out to collect diagnostic samples from aborted materials and aborting dams and questionnaire data on herd/flock management. Laboratory diagnostic tests were carried out for a range of abortigenic pathogens, and laboratory and questionnaire data analysed to investigate a) determinants of attribution; b) patterns of investigated events, including species and breed affected, previous abortion history and recent stressful events, and the seasonality of cases; c) determinants of reporting, investigation and attribution; (d) cases in which zoonotic pathogens were detected.

Results

Over a two-year period (2017-2019), 215 abortion events in cattle (n=71), sheep (n=44) and goats (n=100) were investigated from 13 of the 15 wards. The number of investigated cases varied widely across wards, with three of the 15 field officers (20%) reporting 70% of investigated cases. Aetiological attribution, achieved for 19.5% of cases through PCR-based diagnostics, was significantly affected by delays in obtaining samples from field investigation. Histopathology proved less useful than PCR diagnostics due to rapid deterioration of field samples. Vaginal swabs from aborting dams provided a practical and sensitive source of diagnostic material for pathogen detection.

Discussion / Conclusion

Livestock abortion surveillance, even at a small scale and when capturing only a small proportion of events, can generate valuable information on causes of zoonotic disease outbreaks, livestock reproductive losses and can identify important pathogens that are not easily captured through other forms of livestock disease surveillance. This study demonstrated the feasibility of establishing an effective reporting and investigation system that could be implemented across a range of settings, including remote rural areas. This has been achieved through effective engagement of community-based field officers, establishment of practical sample collection protocols and application of molecular diagnostic platforms.

eLife assessment

This **important** study reports the use of a surveillance approach in identifying emerging diseases, monitoring disease trends, and informing evidence-based interventions in the control and prevention of livestock abortions, as it relates to their public health implications. The data support the **convincing** finding that abortion incidence is higher during the dry season, and occurs more in cross-bred and exotic livestock breeds. Aetiological and epidemiological data can be generated through established protocols for sample collection and laboratory diagnosis. These findings are of potential interest to the fields of veterinary medicine, public health, and epidemiology.

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Introduction

Livestock reproductive losses, including spontaneous abortion, are a major concern for the livestock industry worldwide, resulting in significant economic loss and posing a threat to public health (Food and Agriculture Organization 2011 [↗](#)). The impacts of livestock abortion on the world's poorest livestock-keepers, who are heavily dependent on livestock for food security and livelihoods (Food and Agriculture Organization 2011 [↗](#)), is likely to be substantial. For these families, the loss of a livestock foetus and a subsequent decline in milk production reduces the availability of a high-quality food source (milk) that can be essential for childhood growth and cognitive development (Neumann *et al.* 2003 [↗](#)). In addition, livestock reproductive losses reduce income (from sales of meat and milk), and cause a loss of livestock assets that are a critical source of wealth, collateral or a safety net in times of need (The World Bank 2021). Recent analyses have shown that because of the need for increased spending on animal management, livestock abortions may also have indirect negative impacts on household expenditure and education (Ahmed *et al.* 2019 [↗](#)). Livestock abortion can also pose a direct threat to human health because many abortigenic agents are also zoonotic (Givens and Marley 2008 [↗](#); Thomas *et al.* 2022 [↗](#)).

Surveillance is defined as the real-time (or near real-time) collection, analysis, interpretation, and dissemination of health-related data to enable the early identification of the impact (or absence of impact) of potential health threats, which require effective action (Hulth 2014 [↗](#)). Effective livestock health surveillance provides critical data for evidence-based approaches to livestock disease control and management but this requires reliable, high-quality, and timely data that can be drawn from multiple sources (George *et al.* 2021 [↗](#)). Over the past decade, increasing attention has been given to animal health syndromic surveillance (Dórea and Vial 2016 [↗](#)), which relies on detection of health indicators, such as livestock abortion, that are discernible before a confirmed diagnosis is made. However, systematic reviews of the literature that collectively cover a period from 2000 to 2016 indicate that syndromic surveillance programmes have mostly been implemented in Europe, North America or Australasia with only a single pilot project identified in Africa (Dórea and Vial 2016 [↗](#); Dorea, Sanchez, and Revie 2011 [↗](#)).

In public health, event-based surveillance has also been gaining attention for the early detection of unusual events that might signal acute and emerging human health risks (World Health Organization 2008 [↗](#); Africa Centres for Disease Control and Prevention 2018). This involves the collection, analysis and interpretation of information through both formal and informal channels to rapidly identify unusual or unexpected health events, such as disease outbreaks, emerging infectious diseases, or other public health threats. Although the WHO guidelines for event-based surveillance (World Health Organization 2008 [↗](#)) make explicit reference to the capture of information around unusual disease events in animals, the development of integrated early-warning systems involving animal disease events is still very limited. Livestock surveillance has a clear potential for identifying and preventing outbreaks of zoonotic and emerging diseases; a substantial proportion (39.4%) of livestock pathogens infect humans (Cleaveland *et al.* 2001 [↗](#)), and ungulates harbour more zoonotic pathogens than any other taxonomic group (Woolhouse and Gowtage-Sequeria 2005 [↗](#)). With several abortigenic agents also causing emerging livestock diseases, for example those caused by Schmallenberg virus, bluetongue virus and porcine reproductive and respiratory syndrome virus, livestock abortion may represent a particularly important syndromic target for zoonotic and emerging disease surveillance. However, there is limited information available on current practices, effectiveness, and challenges/opportunities of livestock abortion surveillance, particularly in low- and middle-income countries.

Livestock abortion has many causes, including infectious, nutritional, physical and genetic factors. The diversity of causative agents and variation in the relative importance of agents across different livestock management systems and geographies makes abortion diagnosis complex and

challenging (Wolf-Jäckel *et al.* 2020 [↗](#)). Challenges relate to timely collection of diagnostic samples, sample availability and deterioration, biases in detectability of agents, as well as complexities around establishment of a causal aetiology, particularly for pathogens such as *Coxiella burnetii* and *Escherichia coli*, which can often be present as incidental infections (Thomas *et al.* 2022 [↗](#); Wolf-Jäckel *et al.* 2020 [↗](#)). As a result, aetiological attribution of livestock abortion rarely exceeds 35%, even in well-resourced industrialised livestock systems (Cabell 2007 [↗](#); Wolf-Jäckel *et al.* 2020 [↗](#)) and data are particularly sparse in African livestock systems.

A recent prospective study of the aetiologies of livestock abortion, carried out in northern Tanzania (Thomas *et al.* 2022 [↗](#)), investigated 215 cases of livestock abortion of which an attribution was made in 41 cases (19%). The infectious causes of abortion were identified as Rift Valley fever virus (RVFV) in 14 cases (6.6%), followed by *Neospora caninum* in ten (4.7%), pestiviruses in nine (4.2%), *Coxiella burnetii* in six (2.8%), and *Brucella* sp., and *Toxoplasma gondii* in one case each (0.5%). Our study draws on the operational data generated from establishment of a livestock abortion study to examine characteristics of reporting and investigation of cases of livestock abortion. While this study was not designed as a systematic or comprehensive evaluation of a surveillance system, we present data on several key attributes (ECDC 2014 [↗](#)) of the platform, including simplicity, data quality, representativeness, timeliness and usefulness. We discuss our findings in relation to the feasibility, practicality, and value of establishing a livestock abortion surveillance framework to support evidence-based interventions to improve livestock development, livelihoods, and human health in Africa.

Methods

Logic model

A logic model was created to provide a conceptual framework that described the logical links between the main activities, outputs, and outcomes that were expected from the programme (Figure 1 [↗](#)). The model depicts the overarching assumption that building better livestock abortion surveillance systems and strong community partnerships will lead to data-driven interventions to prevent and control infectious causes of livestock abortion and to catalyse changes in knowledge, attitudes, behaviours, or practices that could improve livestock productivity, livelihoods, and human and animal health.

Abortion surveillance platform

The abortion surveillance platform was set up in northern Tanzania through a collaboration between the Ministry of Livestock and Fisheries, local government authorities, and the research team. The study was undertaken from October 2017 through September 2019 in 15 wards of five districts of Arusha, Kilimanjaro, and Manyara Regions in northern Tanzania (Figure 2 [↗](#)). These study wards were selected from randomly selected wards included in earlier cross-sectional exposure studies (Bodenham *et al.* 2021 [↗](#)). Thirteen wards were selected at random and two additional wards were selected purposively because of strong existing relationships with the livestock-keeping community (Thomas *et al.* 2022 [↗](#)). These 15 wards comprised five wards that were expected to be predominantly pastoral, three were expected to be predominantly agropastoral and seven expected to be predominantly smallholder, with categories assigned by the research team following discussion with local experts (typically the district level veterinary officer) (Bodenham *et al.* 2021 [↗](#)). Recruited livestock field officers (LFOs) responsible for each target ward received training on the causes and safe investigation of livestock abortion. These officers are government employees that are equivalent to para-veterinarians in other settings. They were requested to ask livestock owners to report any incidents of livestock abortion, stillbirths, and peri-natal death (hereafter referred to as abortion cases).

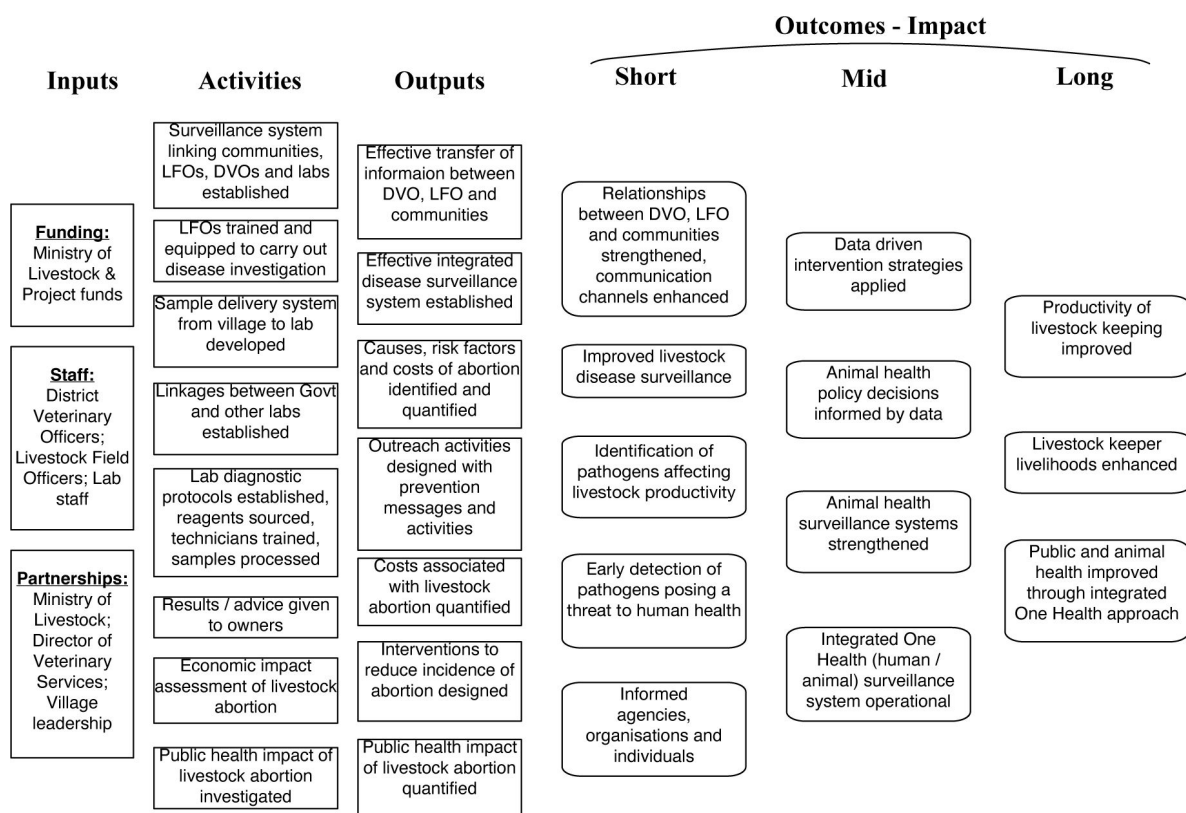


Figure 1.

A logic model illustrating the conceptual links between the inputs, activities, outputs, and short to long term impacts expected from effective livestock health surveillance with a particular focus on abortion.

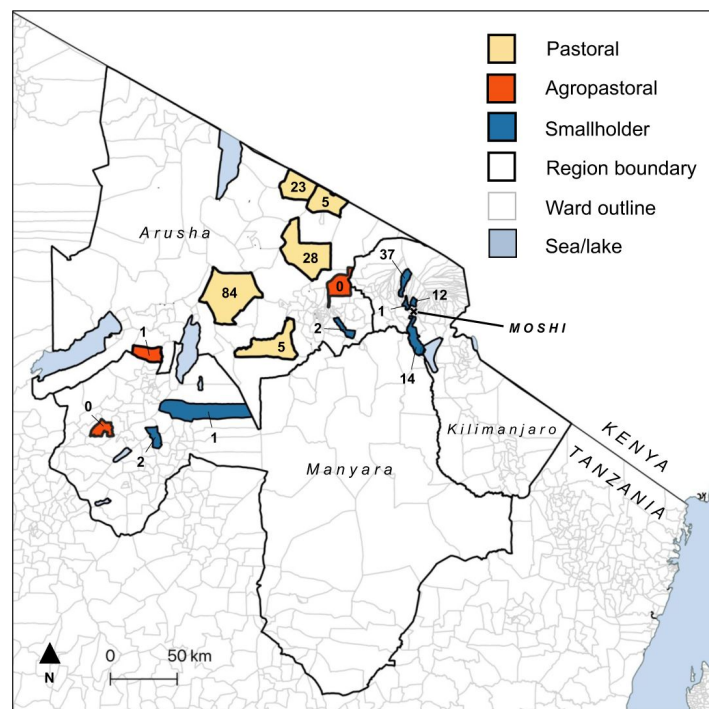


Figure 2

Map of the study area in northern Tanzania showing selected pastoral, agropastoral and smallholder wards in Kilimanjaro, Arusha and Manyara Regions. The number of investigated cases per ward and the study base (Moshi) are shown.

Investigation of cases

Cases were investigated if, following a report from the LFO, the event could be followed up within 72 hours of the abortion occurring. Full details of sample collection are provided in [Thomas *et al.* \(2022\)](#). Briefly, where available, blood, milk and vaginal swabs were collected from the aborting dam and tissue and swab samples collected from the foetus and placental membranes.

Information about the abortion event was collected and a household questionnaire (comprised of mixed open and closed questions) conducted to collect information on livestock demographics, livestock abortion history, the aborting dam (age, breed), household livestock parturition practices and household socio-economic data (Supplementary Material 1). Questionnaire data were only collected from abortion cases that were investigated by the research team and were used to investigate underlying patterns of abortion, risks associated with abortion cases, and operational aspects of the surveillance platform LFOs were instructed to provide farmers with advice as to locally appropriate preventive measures that could be taken to reduce transmission or contamination risks associated with abortion cases, which included safe removal of abortion tissues from livestock-occupied areas (e.g. burning, burying or covering the tissues in thorny branches) (Supplementary Materials 2). Diagnostic results were reported back to LFOs and livestock owners within 10 days of the investigation and, where pathogens were detected, more specific advice provided as to appropriate management strategies that could minimise further transmission to livestock and people.

Event data were collected using a paper-based Cardiff Teleform system (Cardiff Inc., Vista, Ca., USA) into an Access database (Microsoft Corp, Va., USA). Household questionnaire data were collected using handheld digital devices programmed with the Open Data Kit™ survey tool. Data were imported into R (R Core Team 2023) for cleaning, coding and analysis. The survey instruments were pre-tested in wards that were not targets for this study. Geographic co-ordinates from a central point within the household were collected with a handheld GPS (Garmin eTrex™).

Sample analysis

Laboratory diagnostic analyses have previously been described in detail ([Thomas *et al.* 2022](#)). Briefly, samples were tested for: (a) *Brucella* spp., *Coxiella burnetii*, *Chlamydia* spp., and *Leptospira* spp. using quantitative polymerase chain reaction (qPCR) assays; (b) *Neospora* spp. and *Toxoplasma gondii* using conventional nested PCR assays; (c) Bluetongue virus (BTV), bovine viral diarrhoea virus / border disease virus (BVDV/BDV), and Rift Valley fever virus (RVFV) using quantitative reverse transcriptase PCR (RT-qPCR); and (d) additional tests for *C. burnetii* and *Neospora* spp. using immunohistochemistry. Additionally, serum was tested for antibodies to *Brucella* spp., bovine herpesvirus (BHV-1), BVDV, *C. burnetii*, *Leptospira hardjo*, and RVFV.

Summarising the investigated cases and description of the livestock study population

The number of abortion cases was recorded for each agro-ecological zone (pastoral, agro-pastoral and small holder) and study ward, together with the number of households that had an abortion event, the composition of the livestock herds (cattle, goats and sheep) kept at each household, and the number of previous abortion cases in each household that had an event.

The sensitivity of the platform was examined by dividing the number of investigated abortion cases by the *expected* number of abortions for the livestock population of the study wards over the study period. The expected number of abortions was estimated by multiplying (a) the number of abortion cases per head of livestock reported over a 12 month period obtained through a previous randomized cross-sectional study (described in [de Glanville *et al.* 2022](#)) by (b) the number of livestock in the ward reported by surveys conducted by LFOs from the Tanzania Ministry of

Livestock and Fisheries covering a period from 2011 to 2016 (E. Swai, *unpublished data*). This figure was multiplied by two to account for the 24-month duration of the study. Where data were not available for a specific ward, figures were estimated from the average of other wards in the same division (Rau) or by current estimates provided by the LFO (Machame Mashariki) (Supplementary Materials 3).

Determinants of investigation

The determinants of event investigation were analysed to determine the number of cases that each LFO investigated and the distribution of time taken between the report and the subsequent investigation by the research team. The relationship between the time taken between a report and its subsequent investigation and the distance (km) of the ward from the research team headquarters (in the town of Moshi) was investigated using Pearson correlation analysis (**Figure 3** [↗](#)).

Sample collection data

Data were recorded and summarized for: (a) the types of samples collected for each abortion event; (b) the availability of placental and foetal materials and, when not available, the reasons why these materials were not available; and c) the relationship between sample types and pathogen detection and pathogen attribution.

Observed patterns in investigated abortions

The distribution of investigated abortion cases was examined in relation to a) livestock species and breed in which they occurred, b) history of previous abortion cases in the aborting dams, history of recent stressful events affecting the pregnant dam, and d) seasonality of the cases.

1. Breeds were recorded as indigenous (local) and non-indigenous (cross-bred and exotic). We used the breed distribution in the investigated herds to estimate the expected number of abortions if breed type did not affect the likelihood of abortion. Using an exact binomial test, we compared the expected with the observed number of cases.
2. Data on the occurrence of previous abortion cases and recent stressful incidents were obtained from the questionnaire data collected at the time of the investigation.
3. Stressful incidents were recorded as open-ended questions but were prompted by asking about events such as being chased by predators, drought, unusual handling or recent change in diet or grazing habit.
4. Seasonality was investigated in relation to typical 'dry' and 'wet' seasons and in relation to monthly rainfall in the Arusha region during the project period. Mean monthly rainfall was calculated from daily rainfall data, which was obtained across the Arusha Region (Ashouri *et al.* 2015 [↗](#); Sorooshian *et al.* 2019 [↗](#)) and plotted alongside the temporal pattern of reported cases. The relationship between mean monthly rainfall and the number of reported abortions was investigated using linear regression.

All processing and analysis of data was carried out using the statistical programming tool, R (R Core Team 2023 [↗](#)).

Determinants of attribution

World Organization for Animal Health (WOAH – previously OIE) guidelines and case definitions, in conjunction with recommendations from specialist/reference laboratories and peer-reviewed literature, were used to inform diagnosis and attribution of infectious causes of abortion (described in detail in Thomas *et al.* 2022 [↗](#)). Binomial logistic regression analysis was carried out with attribution (yes or no) as the dependent variable. Potential independent explanatory variables, including the delay between the event and its investigation, species, and livestock

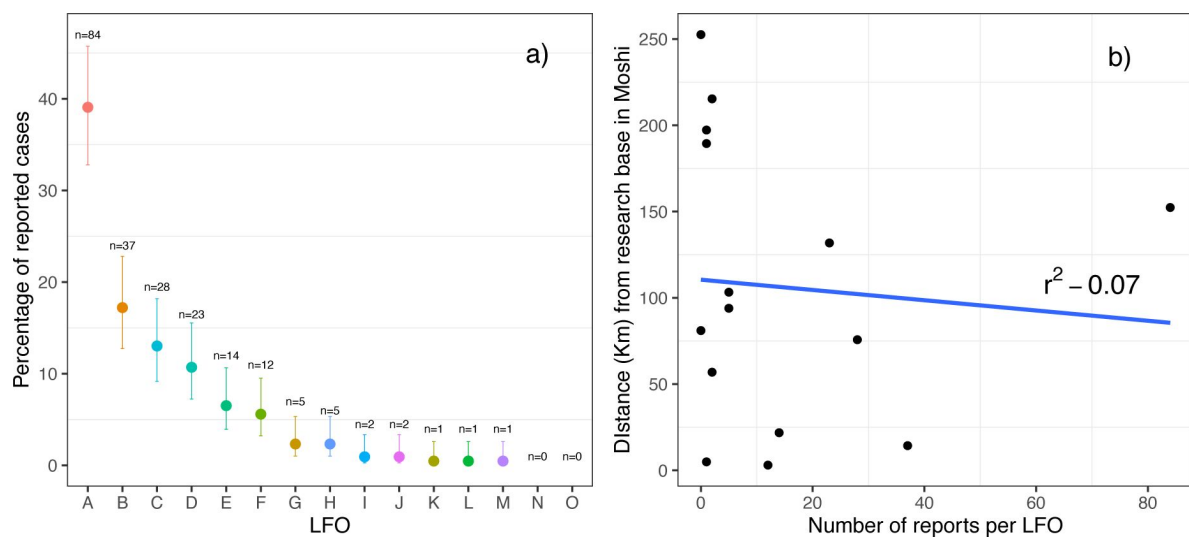


Figure 3

a) The number and percentage of abortion cases reported by each LFO. b) The relationship between the number of reports per LFO and the distance to the research laboratories based in the town of Moshi.

management system, were selected for inclusion in the model through univariable analysis (with all variables with p value < 0.25 selected for inclusion). Potential independent explanatory variables were retained or de-selected through a stepwise approach using Akaike Information Criterion (AIC) as an indicator for model efficiency to achieve a final model.

Exposure to zoonotic pathogens

The percentage of cases in which zoonotic pathogens (*Brucella* spp., *C. burnetii*, *T. gondii* and RVFV) were detected, and the percentage of these cases in which someone assisted with the delivery, was calculated. The age and sex of the persons assisting with delivery was also collected.

Ethical Approval

Ethics approval for this research was granted by Kilimanjaro Christian Medical Centre (KCMC) Ethics Committee (No. 535 and No. 832); National Institute of Medical Research (NIMR), Tanzania (NIMR/HQ/R8a/Vol IX/1522 & NIMR/HQ/R.8a/Vol IX.2028); Research Ethics Coordination Committee and the Institutional Review Board for Clinical Investigations of Duke University Health System in the United States (Pr00037356); University of Otago Ethics Committee (H15/069 & H17/069); University of Glasgow College of Medical Veterinary and Life Sciences Ethics Committee (200140152 & 200170006).

Results

Descriptive Statistics

Between October 2017 and September 2019, 215 abortion cases were reported from 150 households in 13 of the 15 target wards (**Fig. 2**). The distribution of investigated cases in relation to agro-ecological system and herd/flock composition is shown in **Table 1**. Out of the 150 households investigated, most ($n=115$) had only one event investigated. Of the remaining households, 21 had two cases investigated, eight had three, two had four, three had five and one household had 11 cases investigated. Herd level summary statistics have been provided in Supplementary Material 4.

The sensitivity of the platform (the percentage of expected abortion cases that were investigated) ranged from 0%-12.4% for cattle, 0-1.2% for goats and 0-0.3% for sheep. A higher percentage of expected abortions in cattle were reported in smallholder wards (2.7%) than in other wards (0% for agropastoral and 0.31% for pastoral wards), with particularly high reporting in one smallholder ward, Machame Mashariki, close to Moshi town, where 12.4% of expected cattle abortions were investigated.

Over the 12 months preceding a reported abortion event, abortions, peri-natal mortalities, or stillbirths occurred in 52 of 184 (28.3%) cattle herds, 98 of 168 (58.3%) goat herds and 93 of 140 (66.4%) sheep herds that had at least one adult female. Out of herds that had at least one abortion case, the mean (median, range) number of cases of abortion per herd in the past 12 months was 3.7 (3, 1 – 12) for cattle, 9.0 (6, 1 – 70) for goats and 6.7 (4, 1 – 67) for sheep.

Determinants of investigation

The number of cases reported to the research team by each LFO varied considerably (median = 5, range 0 – 84). Of the 215 cases, 70% were reported by three (20%) of the LFOs, with one reporting 84 cases (39.1%). Two LFOs did not report a single case (**Figure 3a**). The range in the interval between the report and the subsequent investigation by the research team was zero to four days with a median of one day (**Figure 4**).

Event	Category	Number (%)
Number of abortion cases	All species	215
	Cattle	71 (33%)
	Goats	100 (46.5%)
	Sheep	44 (20.5%)
Number of abortion cases in each agricultural ecological zone	Pastoral	144 (67.0%)
	Agro-pastoral	1 (0.5%)
	Peri-urban	70 (32.5%)
Number of households that had an abortion event	Households	150
Number of households in each agricultural ecological zone	Pastoral	84 (56.0%)
	Agro-pastoral	1 (0.7%)
	Peri-urban	65 (43.3%)
Composition of the 150 herds	Cattle, goats and sheep	77 (51.3%)
	Cattle and goats	17 (11.3%)
	Goats and sheep	7 (4.7%)
	Cattle and sheep	1 (0.7%)
	Cattle only	40 (26.6%)
	Goats only	8 (5.3%)

Table 1

The number (and percentage) of abortion cases by species and agro-ecological zone and the composition of the livestock herds (and percentage) in investigated households.

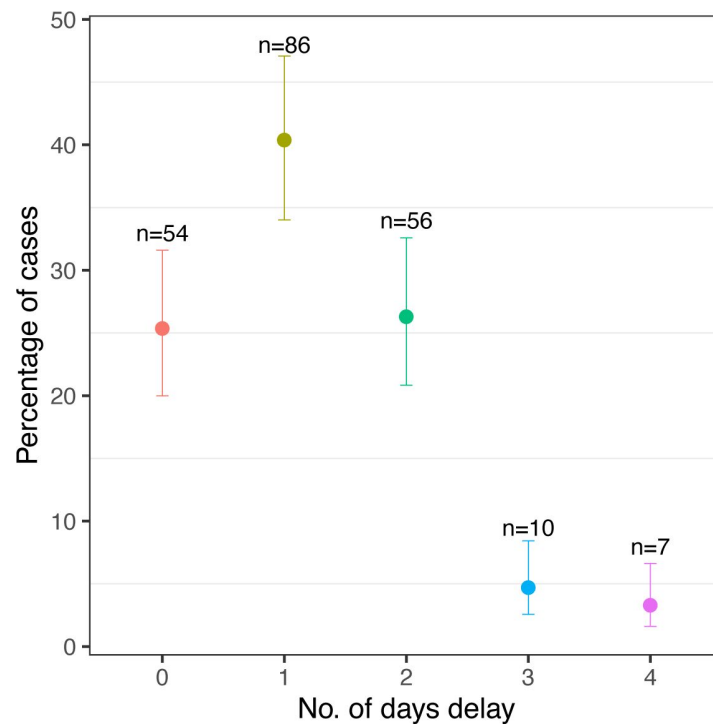


Figure 4

The number of days between abortion and the investigation. No cases were investigated more than after four days after the abortion.

From the Pearson correlation analysis, no significant association was found between the interval between reporting and investigation and the direct distance between the centroid of the ward in which the LFO worked and the research team headquarters ($R^2 = -0.07$, $t = -0.32$, $p = 0.75$) (**Figure 3b** [↗](#)).

Sample collection data

Out of the 215 cases, placental and foetal tissues were collected in 116 (24.0%) and 141 (34.1%), respectively. The reasons given for failure to collect placental and foetal tissues included: a) the tissues not being seen by owners (for example if the animal aborted while away grazing); b) the tissues being burned by the owner; and c) the tissues being consumed by dogs or other animals. Vaginal and milk samples from aborting dams were collected in 213 (99.1%) and 167 (77.7%) cases, respectively.

Observed patterns in investigated abortions

a) Pattern of abortions in species and breeds

In cattle, reported abortions occurred significantly more often than expected in non-indigenous cross-bred animals (expected proportion = 0.11, actual proportion = 0.52, 95% CI: 0.42 – 1.00, $p < 0.001$) and non-indigenous exotic animals (expected proportion = 0.01, actual proportion = 0.25, 95% CI: 0.16 – 1.00, $p < 0.001$). In goats, reported abortions occurred significantly more often than expected in non-indigenous cross-bred animals (expected proportion = 0.02, actual proportion = 0.18, 95% CI: 0.11 – 1.00, $p < 0.001$) and more often than expected in non-indigenous exotic animals, although this difference was not significant (expected proportion = 0.01, actual proportion = 0.03, 95% CI: 0.01 – 1.00, $p = 0.053$) (**Figure 5** [↗](#) and Supplementary Materials 5). There was no significant difference in the distribution of abortions in different breeds of sheep.

b) History of previous abortion cases in the aborting dams

Of the cattle, goat and sheep dams that were investigated in this study (and that had had previous pregnancies), 33.3% ($n=12$), 29.8% ($n=17$) and 16.7% ($n=5$), respectively, were reported by the owner to have experienced a previous abortion event. Of these cattle and goats, 41.7% ($n=5$) and 47.1% ($n=8$), respectively, had suffered multiple previous abortions. In one particular case a cow had experienced four previous abortion cases and one goat had experienced seven.

c) History of recent stressful events

Of dams that aborted, 16 of 71 cattle (22.5%), 37 of 98 goats (37.7%) and 15 of 43 sheep (34.9%) were reported to have experienced recent stress. Dams that aborted for which an attribution was made were no more or less likely to have experienced a stressful event than dams for which an aetiological attribution was not made. Regarding recent illnesses over the previous four weeks (diagnosed by the farmer), cattle were reported to have suffered from a range of conditions including anaplasmosis, diarrhoea, lumpy skin disease, and trypanosomiasis, whilst goats were reported to have suffered predominantly from respiratory disease.

d) Seasonality of cases

Cases of abortion were reported in every month of the 24-month study period, and although these fluctuated over time, with more cases reported during the drier periods (**Figure 6** [↗](#)), there was no significant effect of mean monthly rainfall on the number of cases (correlation coefficient = -0.005 , $t = -0.25$, $p = 0.8$).

Figure 5.

The difference between the expected and actual proportion of abortion cases in each species and breed. Value of 0 = the expected number of cases occurred, > 0 more than expected, < 0 less than expected (LOC= indigenous (local), XB= non-indigenous cross-bred, EX= non-indigenous exotic breed).

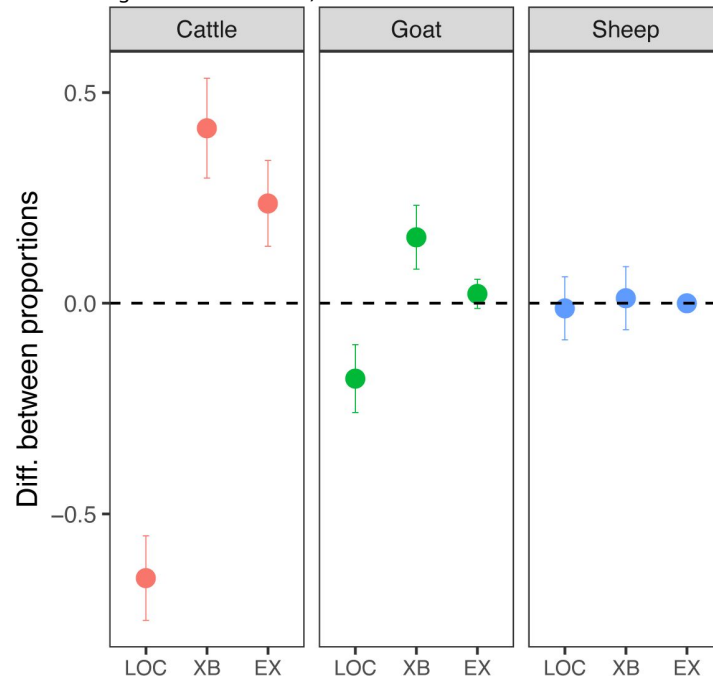
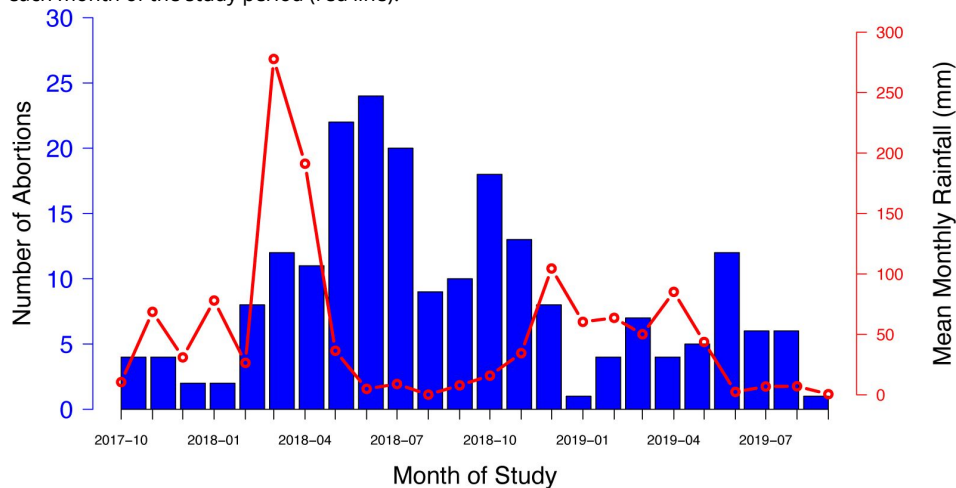


Figure 6.

The number of abortion cases investigated per month (blue columns) shown against mean rainfall recorded in the Arusha region over each month of the study period (red line).



Determinants of Attribution

As described in [Thomas *et al.* \(2022\)](#), the number of cases for which an abortigenic agent was attributed was 42 out of 215 (19.5%). Out of these, an attribution was made using PCR in 41 cases. One event in a cow met the case definition for two pathogens (both BHV-1 through seroconversion and *Neospora* through PCR) and in this event both pathogens were attributed. The attribution of the single case that was not determined using PCR (BHV-1) was made by serology alone. The sample types that were collected in each of these 41 cases for which an attribution was made using PCR, and whether the samples returned a positive or negative result, are shown in [Figure 7](#). The time period between the abortion incident and the investigation (*Delay*) had a negative impact on attribution ($z = -2.1$, $p = 0.03$), with each daily increase in the delay corresponding to a decrease in the odds of an attribution being made by 46.1% (i.e. $1 - 0.539$) ([Table 2](#) and [Figure 8](#)). Additionally, when the abortion occurred in goats (*Goat*) an attribution was significantly less likely than when it occurred in cattle. Finally, the odds of achieving an attribution were not affected by the availability of the placenta (*Placenta present*), foetus (*Foetus present*) or milk (*Milk collected*) for sampling.

Exposure to zoonotic pathogens

Zoonotic pathogens (*Brucella* spp., *Coxiella burnetii*, *Toxoplasma gondii*, and RVFV) were detected in 61 of the 77 (79.2%) abortion cases where a pathogen was detected. Respondents reported that someone had assisted with the delivery in 13 (21.3%) of these cases, similar to the proportion of assisted deliveries across all abortion cases (40 out of 215, 23.5%). Of those assisting with delivery, the median age was 37, the youngest was seven, the eldest 84, and 20% were female.

Discussion

This study demonstrated the feasibility of establishing a surveillance platform for reporting and diagnosing cases of livestock abortion and that these investigations, even at small scale, have generated important information on livestock diseases, and their investigation, in Tanzania. This included information that, to our knowledge, has not been captured or reported through existing surveillance systems.

Key findings were that: i) livestock abortion can be a target for syndromic and/or event-based surveillance, with abortion incidents sufficiently distinctive and noteworthy to be reported by farmers to livestock field officers in a timely manner for investigation; ii) pragmatic and robust protocols for sample collection and laboratory diagnosis can be established, even in resource-limited settings, to generate important etiological and epidemiological data; iii) the likelihood of obtaining an etiological diagnosis depended on the timeliness of reporting and quality of sample collection; iv) there was wide variation in reporting and investigation of cases by different LFOs and this did not appear to be associated with the distance between the ward and the investigation center; v) in many herds prior livestock abortions were reported to have occurred in the 12 months preceding the investigated abortion event; vi) abortions and repeat abortions reported disproportionately more frequently in non-indigenous breeds (cross-bred and exotic) than local breeds; and vii) abortion cases were reported and investigated more frequently during drier periods, although there was no evidence of a relationship with monthly rainfall.

This study suggests that livestock abortion cases are sufficiently distinctive and observable to be reported by farmers to LFOs, would have widespread acceptability as a target for syndromic or event-based surveillance and would have considerable utility for farmers as well as for the livestock and public health sectors. The platform demonstrated the potential for generating high-

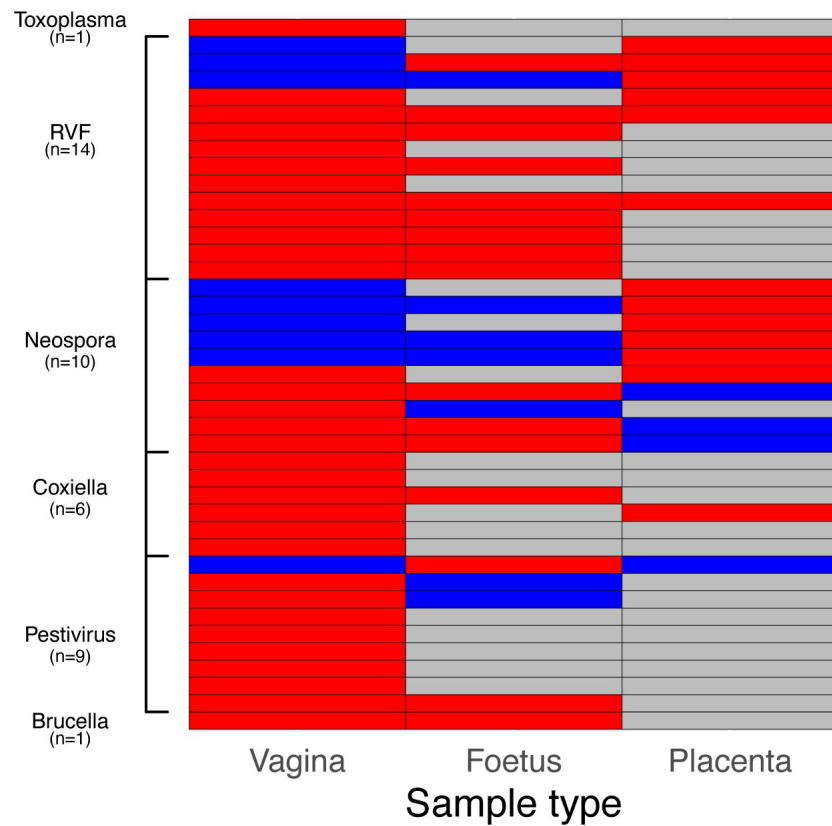


Figure 7

The type of samples in which pathogens were detected in the 41 abortion cases for which an attribution was made using PCR are shown. Each row represents one abortion event. Red - sample type returned a positive result; blue - sample type returned a negative result; grey - sample type was not collected.

Figure 8.

Predicted probability of attribution being made as a function of increasing delay between abortion and case investigation, as determined by the regression model output. The blue line indicates the regression line with the 95% confidence interval shaded blue.

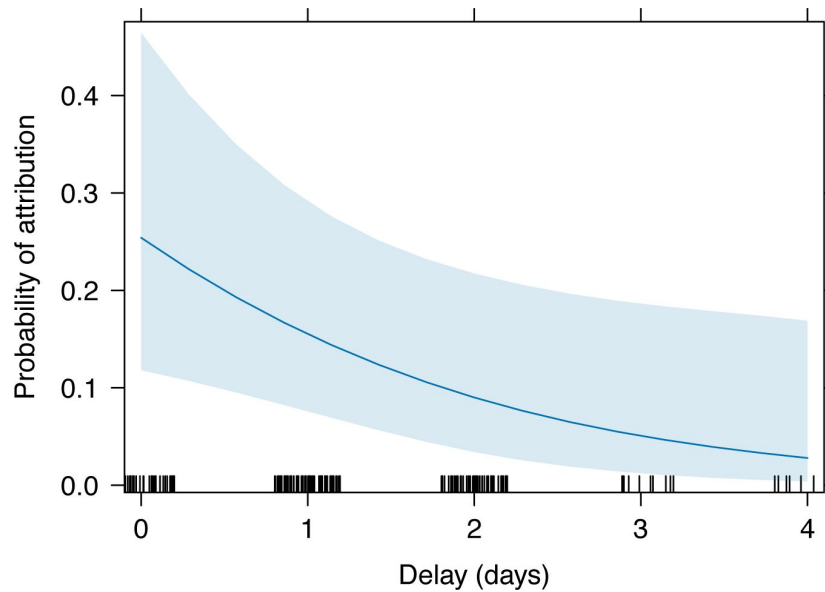


Table 2

Output of final regression model investigating determinants of attribution.

	Odds Ratio	2.5%	97.5%	z value	<i>p</i>
(Intercept)	0.353	0.053	2.361	-1.074	0.283
Delay	0.539	0.306	0.95	-2.139	0.032
Goat	0.195	0.058	0.653	-2.651	0.008
Sheep	0.615	0.142	2.664	-0.65	0.516
Foetus present	0.936	0.345	2.544	-0.129	0.897
Placental present	1.779	0.621	5.094	1.074	0.283
Milk collected	2.405	0.639	9.055	1.297	0.195

quality aetiological data on previously relatively unrecognised livestock disease problems in Tanzania, such as *Neospora caninum* and novel pestiviruses, as well as data on endemic zoonoses, such as Q fever, brucellosis and toxoplasmosis (Thomas *et al.* 2022 [↗](#)). Reports of unusual livestock abortion cases also provided an early warning of acute human health risks, as shown in this study area by the detection of an outbreak of RVF in livestock in peri-urban smallholder cattle (de Glanville *et al.* 2022 [↗](#)), which was the catalyst for subsequent public health investigations that identified several linked human RVF cases, including one fatality (Madut *et al.*, in press).

An attempt was made to gauge the sensitivity of the surveillance platform by estimating the percentage of expected abortions that the investigated cases represented. This was carried out using data on ward-level livestock numbers drawn from Ministry of Livestock and Fisheries records. However, given the dynamic nature of livestock populations in the study area and shifts in ward configurations, these estimates should be considered approximate figures only. Overall the sensitivity of this platform was low, with the highest ward level sensitivity (12.4%) below a mean sensitivity of 34% reported for bovine abortion surveillance systems in France (Bronner *et al.* 2015 [↗](#)). Nonetheless, given that this was the first time that an abortion surveillance platform of this kind had been introduced, and the logistic and communication challenges inherent to this type of low-income setting, our study gives confidence of the feasibility and potential utility of this approach. Despite the limitations of the data, our conclusion that there is wide variation in the sensitivity of the surveillance platform across wards is likely to be robust. This finding, together with the ability of the platform to investigate only a very small proportion of expected abortion cases overall, has implications for the representativeness of data generated, which needs to be considered when interpreting the results reported here.

Contrary to expectations, the location and distance of the ward was not associated with the probability of cases being investigated. Association between cases investigated and farming system could not be determined as the number of wards was relatively small and the number representative of each system were not equal. However, relatively few cases were reported and investigated in the three agropastoral system wards, whilst both high and low rates of reporting occurred in individual wards within the smallholder (seven wards) and pastoral (five wards) systems. This suggests that LFOs and farmers in some wards were much more engaged and motivated than in others. To date, much investment in animal and human surveillance has been directed towards strengthening laboratory diagnostic capacity. While this is clearly essential, support also needs to be given to engaging, training, and motivating farmers and front-line animal health workers to effectively report both routine and unusual animal disease events. Understanding why and how animal keepers and LFOs are incentivised to report and investigate livestock health events will be essential for improving the reach and sensitivity of future surveillance platforms. The infrastructure for LFO engagement across Tanzania provides a valuable platform for timely and cost-effective reporting of abortions and other disease events but seems to be underutilised in existing surveillance systems.

The effectiveness of the platform in allowing an aetiological diagnosis to be reached was significantly impacted by the timeliness of investigation. The impact of delays may be explained by degradation of diagnostic samples, which also affected the utility of histopathological diagnosis (Thomas *et al.* 2022 [↗](#)). Delays were caused by several factors, including the time taken by the investigation team to reach households due to remoteness of some areas and multiple cases being reported in different areas on the same day. These findings underscore the importance of being able to respond rapidly to cases and, where access is difficult, of providing locally suitable means of transport.

A reactive surveillance platform that is managed locally, for example by ward-based LFOs, will require the establishment of effective and safe protocols for collection and transport of samples for laboratory diagnosis that can be carried out without the need for long-distance travel of more highly trained investigators. Protocols for collection of vaginal swabs from aborting dams, which

proved effective for both pathogen detection and attribution, may be of value. In this study, the greater accessibility of vaginal swabs, as compared to placental samples and foetal samples, meant more attribution of abortigenic pathogens was attained through these samples (**Figure 7**). They also require less handling of potentially infectious tissues and are more reliably accessible than placental and foetal tissues, which are often consumed by scavengers or disposed of by the farmer. Indeed, collection of vaginal swabs from the dam was possible in almost all cases investigated in this study. Thus, while logistic, financial, and capacity constraints for comprehensive sampling and investigation of livestock abortions are likely to exist across Africa, these need not preclude the establishment of simple and robust protocols that can yield valuable surveillance data.

Attribution

From this platform, an attribution was reached in 19.5% of cases (Thomas *et al.* 2022), which was not far below the typical range of 25–45% achieved in industrialized farming systems in high-income settings (Campero *et al.* 2003; Amouei *et al.* 2019; Derdour *et al.* 2017; Anderson, Barr, and Hoffman 1990). During this study, samples were tested for only ten abortigenic pathogens and it is likely that rates of detection and attribution would be higher with inclusion of tests for other known abortigenic agents (such as *Listeria* spp., *Campylobacter* spp., *Salmonella* spp. and fungal pathogens), with more locally relevant aetiological data informing the suite of pathogens to be tested, and with metagenomic approaches.

As elsewhere in the world, non-infectious conditions, including nutritional, metabolic, and toxic conditions (Alemayehu *et al.* 2021; Mee *et al.* 2023; Woodburn *et al.* 2021), are also likely to cause livestock abortions in Tanzania. Our study suggests a possible role for recent stressful events, including episodes of drought and attacks by wild predators, which were reported from approximately a third of unattributed abortion cases. However, these events are a common occurrence in our study population, so attribution is likely to be challenging.

We do not have an immediate explanation as to why the likelihood of attribution was lower for goats than for sheep or cattle. One possible explanation is that, in comparison to sheep and cattle, fewer data are available worldwide on causes of abortions in goats and very little is known about the aetiology of goat abortions in Africa, so the pathogens included in our diagnostic panel may not have been as relevant for goats as for sheep and cattle. This lack of knowledge is highlighted by the unusual finding of *Neospora caninum* as a cause of abortion in a goat detected through this platform (Thomas *et al.* 2022).

The challenges of attribution, discussed also in Thomas *et al.* (2022), should not detract from the value of reporting and investigating livestock abortion cases. Recent studies have indicated that the economic costs of livestock abortion in Tanzania and impacts on food security are much more substantial than previously recognized; for example Semango *et al.* 2024 estimated the annual gross losses associated with abortion in Tanzania to be \$262 million USD. Given these findings, data collected on the number of cases and species/breed affected will be of considerable value in highlighting the importance of this syndrome in the context of livestock productivity, household livelihoods and food security.

Patterns of livestock abortion in northern Tanzania

Data from the household questionnaire indicated that livestock abortion was a common occurrence across the study area with almost a third of cattle herds and two-thirds of small ruminant herds having experienced a previous case of abortion in the preceding twelve months.

This study indicated that investigated cases occurred more often than expected in non-indigenous breeds (cross-bred and exotic cattle and cross-bred goats) than in local breeds. This is consistent with studies reporting a four-times higher risk of abortion in cross-bred cattle in Ethiopia in comparison to local breeds (Deresa, Tulu, and Deressa 2020) and studies in Nigeria (Yakubu,

Awuje, and Omeje 2015 [↗](#)) and India (Khan, Manoj, and Pramod 2016 [↗](#)) reporting breed effects in relation to cattle abortion. Our results may simply reflect a higher likelihood of reporting of abortion in animals that are of higher value and more likely to be zero-grazed or tethered while grazing (and hence abortions would be more apparent) and for which an existing relationship with animal health services is more likely.

More than a fifth of the dams that aborted were reported to have experienced a previous abortion, and several had experienced multiple abortion losses, with non-indigenous breeds more likely than indigenous breeds to have had a previous abortion. Chronic repeat abortions can arise from infection with pathogens such as *Neospora caninum* and pestiviruses which were detected as common causes of abortion in this study. Repeat aborting dams could be identified as candidates for slaughter with potential benefits for herd productivity, as well as reduced risks of potential zoonotic transmission. However, any advice around removal of these animals from a herd may result in sale rather than slaughter, with the risk of further spread of the pathogen.

The risk of exposure to zoonotic pathogens through abortion cases was underscored by two findings: first that, in nearly 80% of cases in which a pathogen was detected, the pathogen was zoonotic and second, that in nearly a quarter of all cases, someone had assisted with the aborted delivery, very likely without any form of personal protective equipment. Of those who assisted, 20% were female, of which most were of reproductive age and therefore at particular risk following infection with pathogens such as *Toxoplasma gondii*.

In conclusion, this study has demonstrated that livestock abortion surveillance, even at a relatively small scale, can capture valuable aetiological and epidemiological information on important livestock pathogens, including those that are zoonotic and those with epidemic potential. The study demonstrated the utility, acceptability, and feasibility of livestock abortion as a target for both syndromic and event-based surveillance and showed that an effective reporting and investigation system could be operationalized across a range of settings in Tanzania, including remote rural areas. Key elements of effectiveness were high levels of engagement of community-based field officers, the establishment of practical and robust field sample collection and application of molecular diagnostic techniques, with prompt response to reporting of cases, and timely feedback of results. Future research might include abortion surveillance implemented at a larger scale to better understand the extent, and economic impact, of livestock abortion and associations with different livestock keeping systems. This work might include the investigation of rapid point-of-care testing and the use of mobile phone technology to speed up the detection and reporting back of results to LFOs and livestock keepers, with the potential to facilitate the capture of data for use in national disease surveillance databases. Such research might allow for refined assessment of the economic impact of abortion and, subsequently, of draftpreventative interventions. Further, our results suggest a possible link between abortion and low rainfall, and further work is needed to examine this potentially important relationship in detail. Additionally, investigating the effectiveness of syndromic or event-based surveillance more broadly, targeting syndromes other than abortion, would provide further evidence demonstrating whether this type of surveillance can be implemented, in a cost-effective manner, at scale. Such research would provide much needed evidence to inform the design of effective livestock health interventions to improve livelihoods, food security and public health.

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Editors

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Reviewer #1 (Public review):

Summary:

The paper examined livestock abortion, as it is an important disease syndrome that affects productivity and livestock economies. If livestock abortion remains unexamined it poses risks to public health.

Several pathogens are associated with livestock abortions but across Africa however the livestock disease surveillance data rarely include information from abortion events, little is known about the aetiology and impacts of livestock abortions, and data are not available to inform prioritisation of disease interventions. Therefore the current study seeks to examine the issue in detail and proposes some solutions.

The study took place in 15 wards in northern Tanzania spanning pastoral, agropastoral and smallholder agro-ecological systems. The key objective is to investigate the causes and impacts of livestock abortion.

The data collection system was set up such that farmers reported abortion cases to the field officers of the Ministry of Livestock and Fisheries livestock. The reports were made to the investigation teams. The team only included abortion of those that the livestock field officers could attend to within 72 hours of the event occurring.

Also a field investigation was carried out to collect diagnostic samples from aborted materials. In addition aborting dams and questionnaires were administer to collect data on herd/flock management. Laboratory diagnostic tests were carried out for a range of abortigenic pathogens

Over the period of the study 215 abortion events in cattle (n=71), sheep (n=44) and goats (n=100) were investigated. In all 49 investigated cases varied widely across wards, with three .The Aetiological attribution, achieved for 19.5% of cases through PCR-based diagnostics, was significantly affected by delays in field investigation.

The result also revealed that vaginal swabs from aborting dams provided a practical and sensitive source of diagnostic material for pathogen detection.

Livestock abortion surveillance can generate valuable information on causes of zoonotic disease outbreaks, and livestock reproductive losses and can identify important pathogens that are not easily captured through other forms of livestock disease surveillance. The study demonstrated the feasibility of establishing an effective reporting and investigation system that could be implemented across a range of settings, including remote rural areas,

Strengths:

The paper combines both science and socio economic methodology to achieve the aim of the study.

The methodology was well presented and the sequence was great. The authors explain where and how the data was collected. Figure 2 was used to describe the study area which was excellently done. The section on Investigation of cases was well written. The sample analysis was also well written. The authors devoted a section to summarizing the investigated cases and description of the livestock 221-study population. The logic model has been well presented

Weaknesses:

All the weaknesses identified have been resolved by the the authors

<https://doi.org/10.7554/eLife.95296.2.sa3>

Reviewer #2 (Public review):

The paper provides a comprehensive analysis of the importance of livestock abortion surveillance in Tanzania. The authors aim to highlight the significance of this surveillance system in identifying disease priorities and guiding interventions to mitigate the impact of livestock abortions on both animal and human health.

Summary:

The paper begins by discussing the context of livestock farming in Tanzania and the significant economic and social impact of livestock abortions. The authors then present a detailed overview of the livestock abortion surveillance system in Tanzania, including its objectives, methods, and data collection process. They analyze the data collected from this surveillance system over a specific period to identify the major causes of livestock abortions and assess their public health implications.

Evaluation:

Overall, this paper provides valuable insights into the importance of livestock abortion surveillance as a tool for disease prioritization and intervention planning in Tanzania. The authors effectively demonstrate the utility of this surveillance system in identifying emerging diseases, monitoring disease trends, and informing evidence-based interventions to control and prevent livestock abortions.

Strengths:

- (1) Clear Objective: The paper clearly articulates its objective of highlighting the value of livestock abortion surveillance in Tanzania.
- (2) Comprehensive Analysis: The authors provide a thorough analysis of the surveillance system, including its methodology, data collection process, and findings as seen in the supplementary files.
- (3) Practical Implications: The paper discusses the practical implications of the surveillance system for disease control and public health interventions in Tanzania.
- (4) Well-Structured: The paper is well-organized, with clear sections and subheadings that facilitate understanding and navigation.

All suggestions made for improvement of the manuscript have been appropriately effected.

Final Recommendation:

Overall, this paper makes a significant contribution to the literature on livestock abortion surveillance and its implications for disease control in Tanzania.

<https://doi.org/10.7554/eLife.95296.2.sa2>

Reviewer #3 (Public review):

The authors delved into an important aspect of abortifacient diseases of livestock in Tanzania. The thoughts of the authors on the topic and its significance have been clarified. The number of wards in the study area, statistical selection of wards, type of questionnaire ie open or close ended. and statistical analyses of outcomes have been clearly elucidated in the manuscript. The exclusion criteria for two wards out of the fifteen wards mentioned in the text are clearly stated. Observations were from pastoral, agro-pastoral and small holder agro ecological farmers. Sample numbers or questionnaires attributed to the above farming systems correlate findings with management systems. The impacts of the research investigation output are clearly visible as to warrant intervention methods. The identified pathogens from laboratory investigation, particularly with the use of culture and PCR, as well as the zoonotic pathogens encountered are stated in the manuscript and the supplementary files.

In conclusion, based on the intent of the authors and content of this research, and the weight of the research topic, the seeming weaknesses in the critical data analysis observed have been clarified, to demonstrate cause, effect and impact.

The authors have carried out the necessary corrections.

The findings do imply that identification of some of the abortifacient of livestock in Tanzania will necessitate important interventions in the control of the diseases in the study area

<https://doi.org/10.7554/eLife.95296.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer 1:

Lines 43 to 46 cannot be referred to as methodology:

"to investigate a) determinants of attribution; b) patterns of investigated events, including species and breed affected, history of previous abortion and recent stressful events, and the seasonality of cases; c) determinants of reporting, investigation and attribution; (d) cases in which zoonotic pathogens were detection".

The above should be deleted from the methodology.

The text is in the abstract and describes, in brief, analyses that we performed and the rationale for these analyses, which we consider relevant for understanding the approach. As such, we think the text should remain.

Italicize et al. in the citations

This has been done.

Reviewer 2:

Data Presentation: While the analysis is comprehensive, the presentation of data could be enhanced with the use of more visual aids such as tables, graphs, or charts to illustrate key findings.

While further visualisation of findings would be possible, we consider the key results are captured effectively in the existing figures and tables. Open access to the data also allows for further analyses that might be of interest to readers.

Discussion Section: The paper could benefit from a more in-depth discussion of the implications of the findings for disease control strategies and policy formulation in Tanzania.

We thank the reviewer for this important comment. In most of the paragraphs of the Discussion we discuss the implications of the findings with specific reference, where relevant, to disease control in Tanzania. For example, in the paragraph regarding human capacity building, we discuss how LFOs might be incentivised to report health events and how this could improve the reach and sensitivity of future surveillance platforms. Similarly, these issues are discussed in other paragraphs of the Discussion.

Future Directions: Including recommendations for future research or areas for further investigation would add depth to the paper.

This suggestion has been acted upon and we have added text in the conclusion to describe recommendations for future research.

Reviewer 3:

The thoughts of the authors on the topic and its significance are implied, and the methodological approach needs further clarity. The number of wards in the study area, statistical selection of wards, type of questionnaire ie open or close-ended. Statistical analyses of outcomes were not clearly elucidated in the manuscript.

The number of wards and how they were selected (from randomly selected wards included in earlier cross-sectional exposure studies (Bodenham et al. 2021)) is described in the Abortion Surveillance Platform section of the Methods. We have added description of the questionnaire to indicate that it was a mixture of open and closed questions. We have reviewed the statistical analyses and consider that they have been fully and appropriately described and so have not changed this.

Fifteen wards were mentioned in the text but 13 used what were the exclusion criteria.

As described, the study focussed on fifteen wards however two wards did not report any cases. As such, investigations only took place in thirteen of the fifteen wards and this has been described in the text.

Observations were from pastoral, agropastoral, and smallholder agroecological farmers. No sample numbers or questionnaires were attributed to the above farming systems to correlate findings with management systems.

As described, the 15 wards comprised five wards that were expected to be predominantly pastoral, three were expected to be predominantly agropastoral and seven expected to be predominantly smallholder, and these categories were assigned by the research team

following discussion with local experts (typically the district level veterinary officer) (Bodenham et al. 2021). As such, we consider this to be described sufficiently.

The impacts of the research investigation output are not clearly visible as to warrant intervention methods.

The aim of this paper was to provide insights on the feasibility and value of establishing a livestock abortion surveillance platform. The aetiological data that could be used to inform specific disease control measures or interventions was the focus of a previous paper (Thomas et al. 2022) as described in the text.

What were the identified pathogens from laboratory investigation, particularly with the use of culture and PCR not even mentioning the zoonotic pathogens encountered if any?

An earlier published paper describing the aetiology of the cases was mentioned (Thomas et al. 2022). This paper fully describes the identified pathogens and the methods used for identification and attribution. Additionally, in the Sample Analysis section we describe the pathogens that were tested and the methods used. In the section Exposure to Zoonotic Pathogens we specifically list *Brucella* spp. *C. burnetii*, *T. gondii* and RVFV and so we consider that we have sufficiently described the pathogens tested for, the methods and the zoonotic pathogens detected.

The public health importance of any of the abortifacient agents was not highlighted.

The Introduction provides background information on the public health importance of abortifacient agents and we dedicate a whole section (Exposure to Zoonotic Pathogens) to the public health implications of the number of cases in which zoonotic pathogens were detected. Additionally, we discuss the implications of this in the Discussion.

Comments in manuscript itself:

Line 230: Why are you estimating. The study was supposed to be based on real time abortion events or at least abortion events within 72 hours

We were estimating the sensitivity of the platform by dividing the number of investigated abortion cases by the number of abortions for the livestock population in each of the study wards that would have been expected over the study period. Because the denominator in this calculation was an expected number, and not a measured count, we can only estimate.

236: In areas where there was no reported abortion event why will you estimate. This action will lead to false conclusion of abortion event in area that did have an event.

We think there has been some misunderstanding of what this section of text was describing. We were not attributing a case to an area where there was none. Rather, as mentioned above, the aim of this particular analysis was to estimate the sensitivity of the platform. To achieve this, we needed to estimate what the expected number of abortion cases in each ward would have been.

279: Give a brief description of R

A citation and some explanatory text have been added.

348: Table 1: Your table did not show cases where estimate values were used

We think this comment has resulted from the confusion described above regarding estimated cases. Table 1 has summary data for the actual cases that were reported in the study and does not have the data for the estimated number of abortions that were expected to have occurred in each ward. As described in line 247, this data is given in Supplementary Materials 3.

404: *Not clear, please rephrase*

This sentence has been re-drafted to improve clarity

467: *Why are you numbering the findings of your investigation in your discussion? You have not told us about the previous abortion event in your study area prior to this study and why you embarked on this study in this regions. The current abortion event situation in your country based on other researchers work is missing and how your findings is important as it related to similar investigation elsewhere.*

We number the key findings for clarity and to make each finding distinct and so prefer to retain it.

The study area was chosen because it was the site of an earlier cross-sectional exposure study within which the wards were randomly selected. As a result, thirteen of the fifteen wards targeted in the reported study were randomly selected. Two additional wards were selected purposively because of strong existing relationships with the livestock-keeping community. This was explained in the Methods in Lines 161 – 164.

Regarding livestock abortion in Tanzania, as explained in the Introduction (lines 112-114), there is little data on abortion in livestock in Tanzania and elsewhere. Nonetheless, in the Discussion, we do describe the results with respect to other abortion studies carried out in

Ethiopia, Nigeria and India (lines 592-598). Moreover, as described in the Introduction (line 90-94), the implementation of syndromic or event-based surveillance in livestock is rare and to the authors' knowledge has mostly been implemented in Europe, North America or Australasia with only a single pilot project identified in Africa.

494: *Why will you use an estimate for abortion event that were not reported*

As described above, this comment reflects a misunderstanding of what was being described. As written in line 494, an attempt was made to gauge the sensitivity of the surveillance platform by estimating the percentage of expected abortions that the investigated cases represented. That is, to estimate the percentage of abortions that the surveillance platform managed to detect, we divided the number of investigated abortions by the expected number of abortions (in each ward). The method for this estimation was described in lines 228-238.

511: *Why was farming pattern excluded. Livestock rearing condition is equally critical for this type of investigation example an animal reared intensive system farming method will definitely experience different stress than livestock on nomadic free range system*

We agree with the reviewer that livestock rearing system might be expected to impact both the aetiology and incidence of livestock abortion. However, because the number of wards was small and the distribution across system not equal, any association between investigated cases and and livestock rearing system could not be assessed. We have made this clearer with additional text in the same paragraph of the Discussion.

529: *Nothing was mentioned about educating the farmers or livestock owners to assist in some instances on possible sample collection during this abortion events and*

sending these samples as quickly as possible to the central laboratory in suitable condition for investigation and result of the finding communicated back to the farmers

Because abortions can be caused by zoonotic pathogens, we did not involve livestock keepers in the collection of samples. Rather, sample collection was carried out by the research team and livestock field officers who had received appropriate training. In addition, results were reported back to the livestock keepers within 10 days of the investigation and, where pathogens were detected, more specific advice provided as to management strategies that could minimise further transmission to livestock and people. This is all described in the Methods (lines 181-199).

| 540: *The livestock owner can be taught how to collect vaginal swab and send samples under suitable condition to the laboratory and the findings reported back to them.*

Please see above response.

| 549: *Please summarise.*

Line 549-581 succinctly describes the attribution of cases to specific pathogens. The text given is required for comprehension and any further summarisation could impact understanding. Consequently, we have left the text as it is.

| 584: *Please summarise.*

Line 584-626 describes the patterns of livestock abortion in Tanzania. The text given is required to fully discuss the findings and any further reduction in text could impact understanding. Consequently, we have left the text as it is.

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